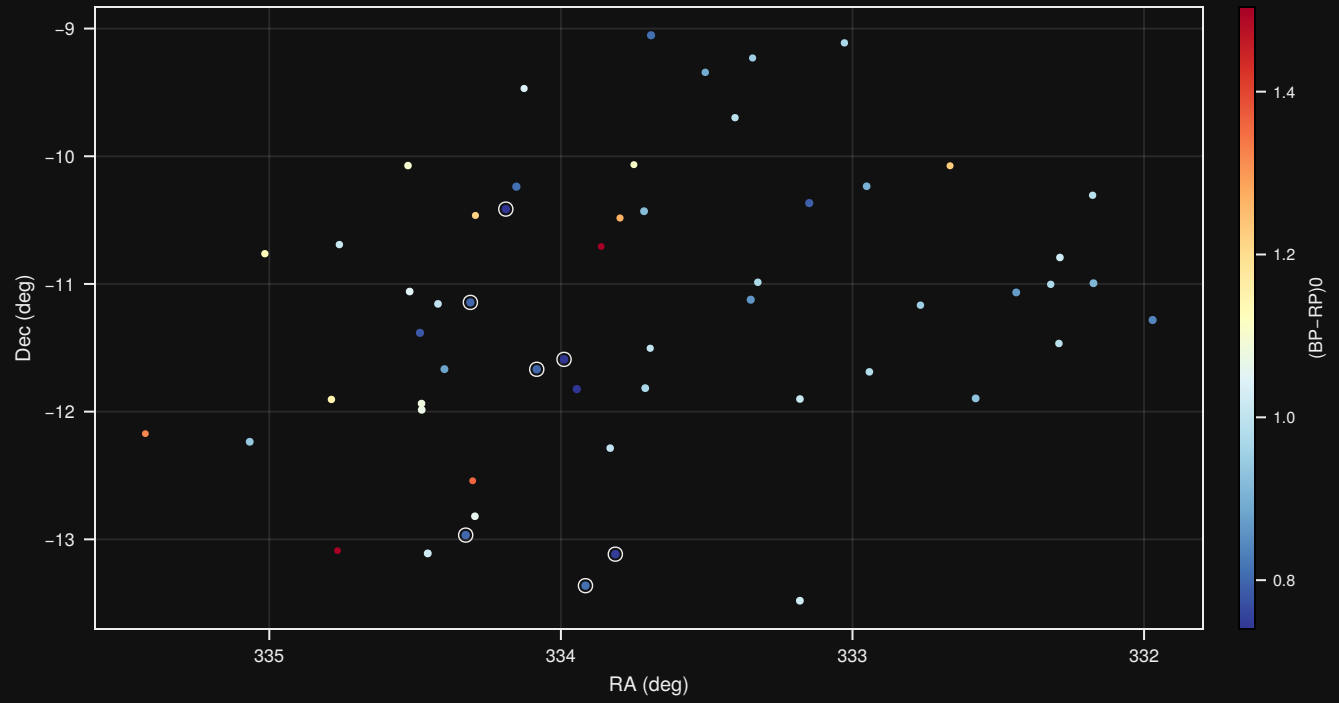
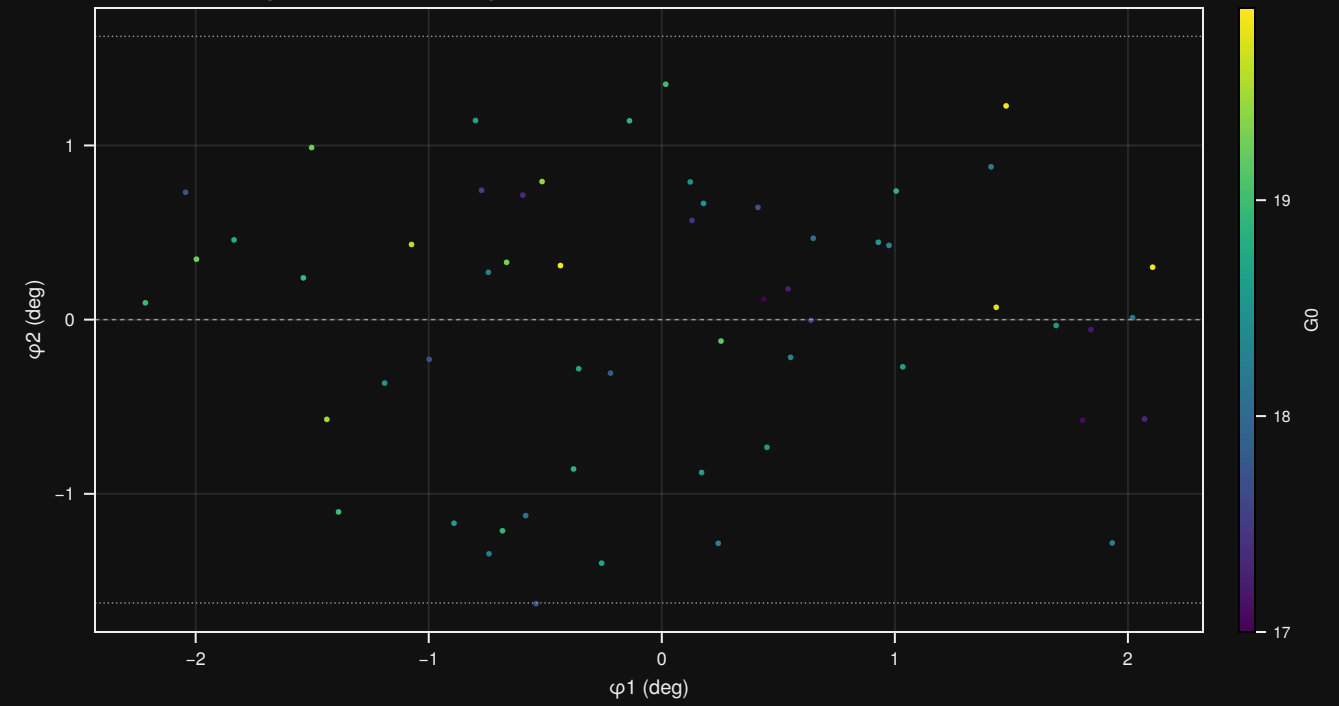
New-24

MEMBERSHIP	
Stars in catalog	56
with Gaia GSP-Phot	40 (71%)
with radial velocity	7
Stream id (sID)	80
GEOMETRY · great-circle fit	
Centroid RA, Dec	333.7, -11.2 deg
Centroid l, b	48.6, -50.0 deg
Pole RA, Dec	59.6, 20.1 deg
Length (φ1, 2–98%)	4.1 deg
Width σ(φ2), robust	0.81 deg
KINEMATICS	
pmRA* med ± σ	1.28 ± 0.64 mas/yr
pmDE med ± σ	-4.70 ± 0.85 mas/yr
μφ1* med ± σ	5.01 ± 0.80 mas/yr
μφ2 med ± σ	-0.25 ± 0.77 mas/yr
Parallax (cat.) med	0.291 ± 0.190 mas
Parallax (wtd, ZP-corr.)	0.318 ± 0.026 mas
Distance	d = 3.1 ± 0.3 kpc
Physical length	0.23 kpc
RV median ± σ(MAD)	-37.5 ± 26.5 km/s
RV gradient dV/dφ1	34.25 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	17.0–19.9 (18.5)
(BP−RP)0 median	0.97
GSP-Phot [M/H] (n=40)	-1.28 ± 0.37
GSP-Phot Teff median	5067 K
RUWE > 1.4	0.0 %
PM clusters (largest)	3 (21 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	56 | 56
n_RV catalog | paper	7 | 7
<[Fe/H]> paper	-0.64
Stellar mass (paper)	170 Msun
Pole offset vs paper	0.6 deg
Remark	Compact, velocities unclear

Sky map · catalog members (n = 56) · size ∝ brightness · ring = has RV

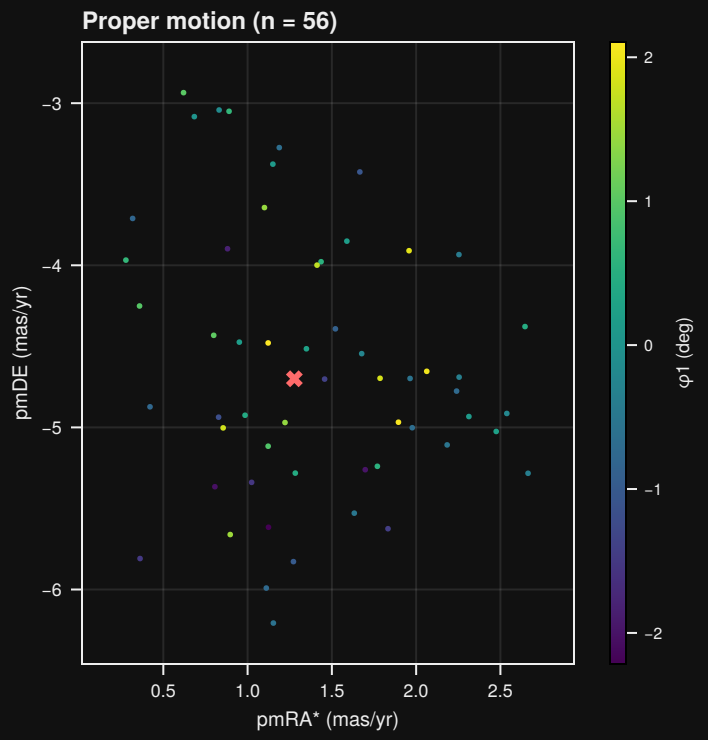


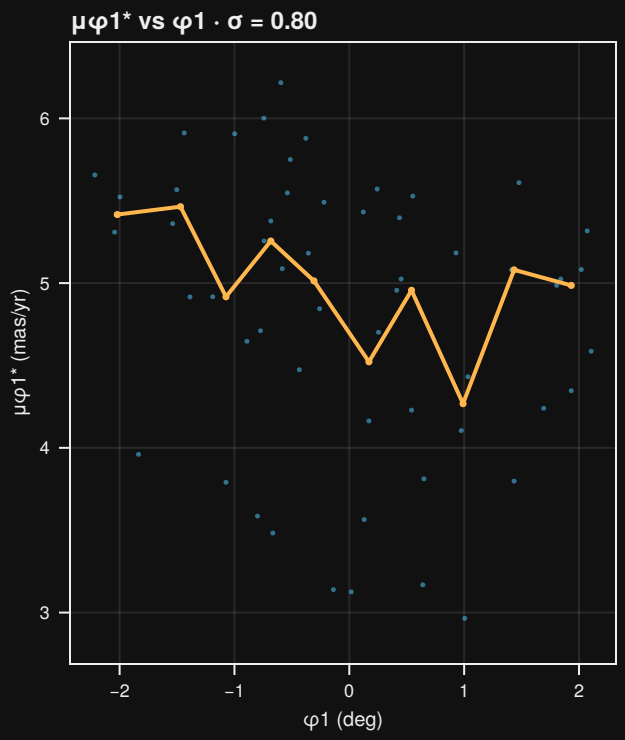
Stream frame (great-circle fit) · length 4.1° · width σ 0.81°

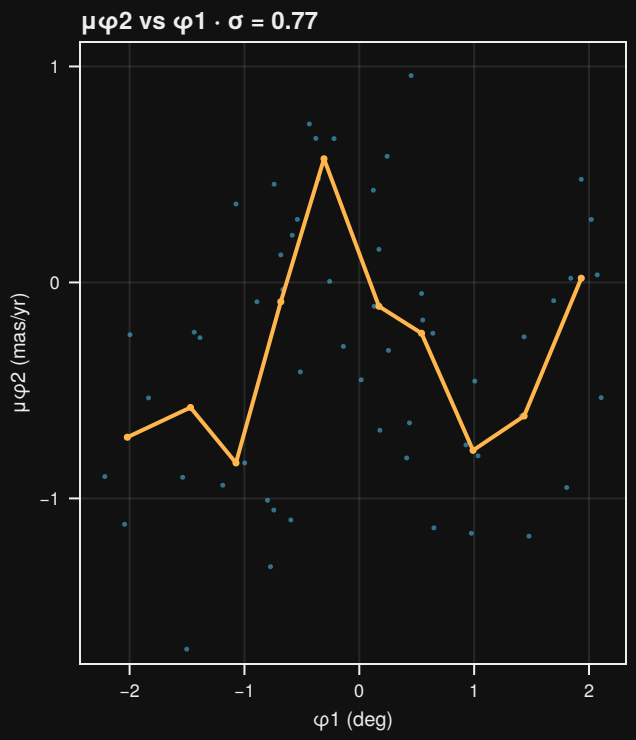


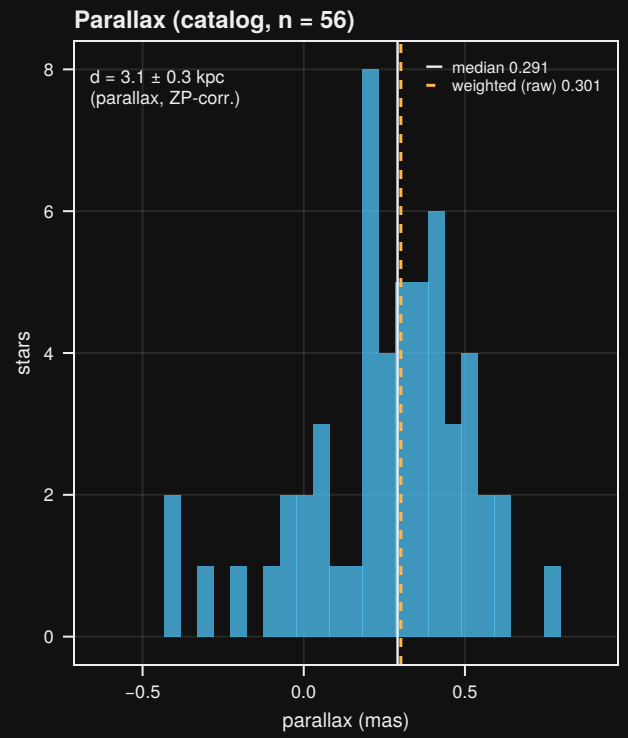
New-24

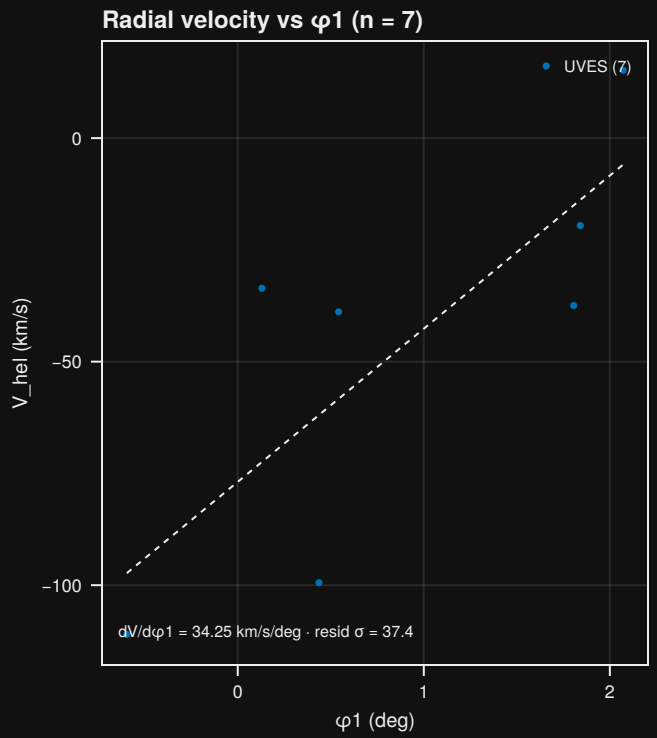
Kinematics · proper motion · parallax · radial velocity

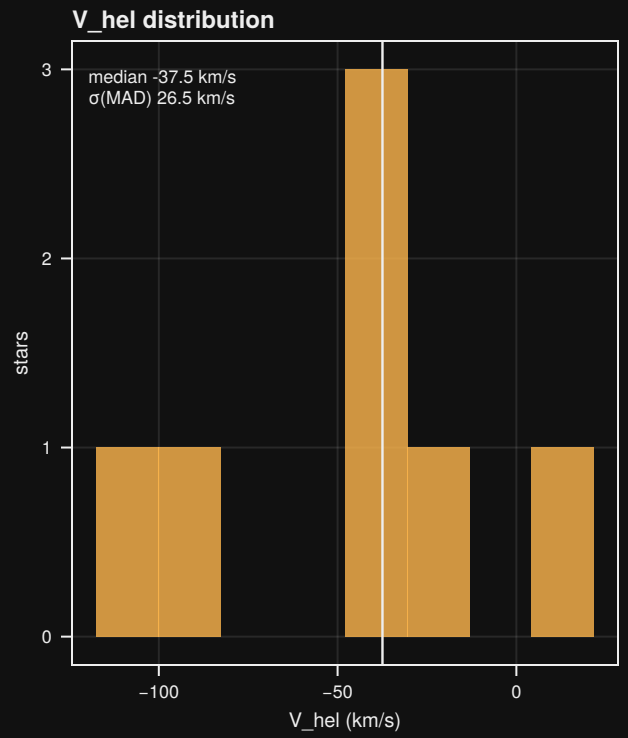








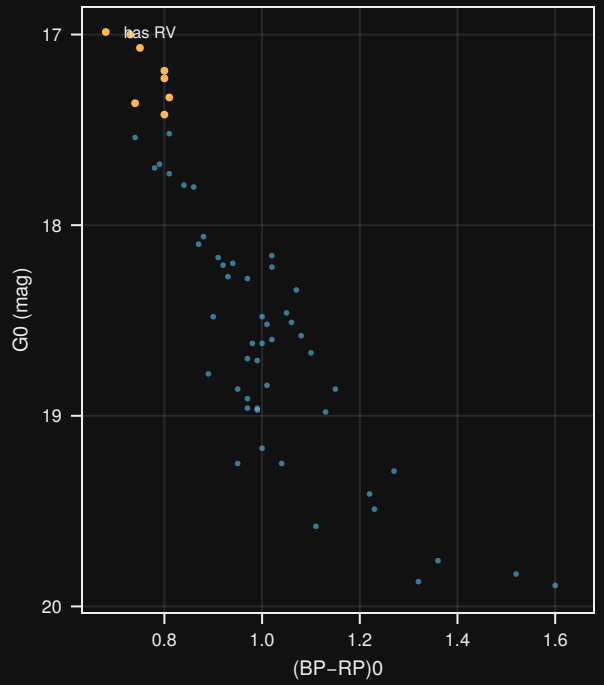




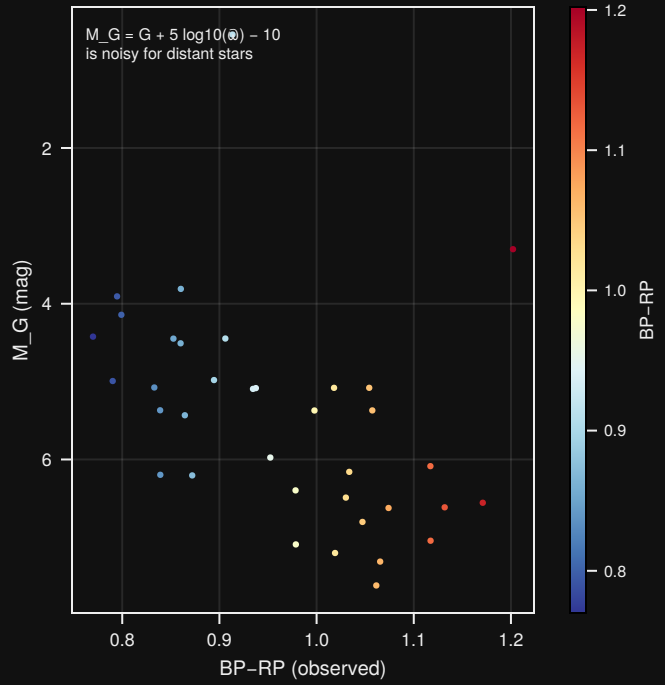
New-24

Photometry · stellar types

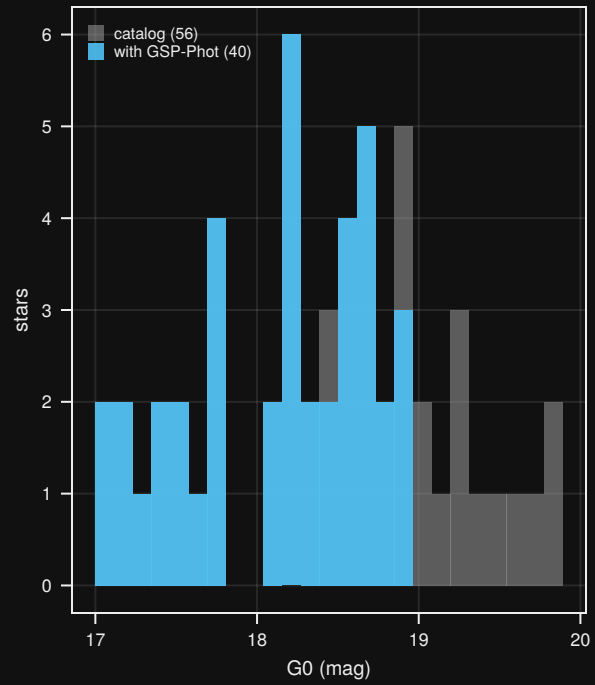
CMD · dereddened (n = 56)



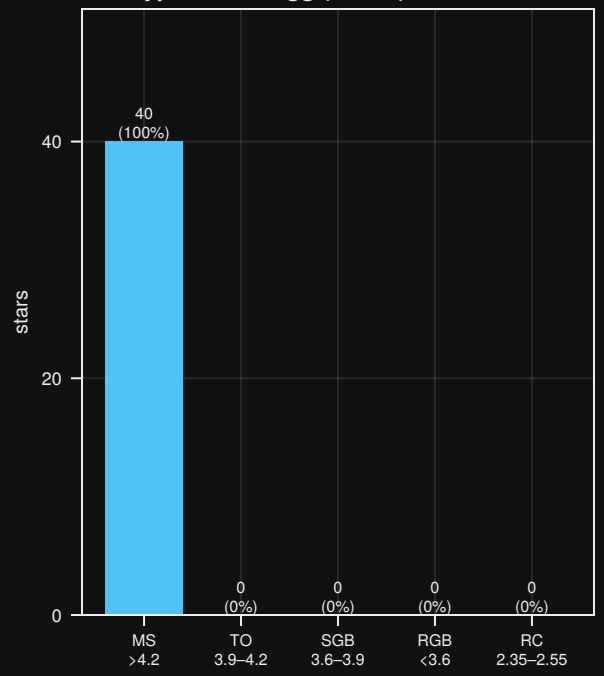
HR diagram · ϖ̃>0 (n = 36/40)



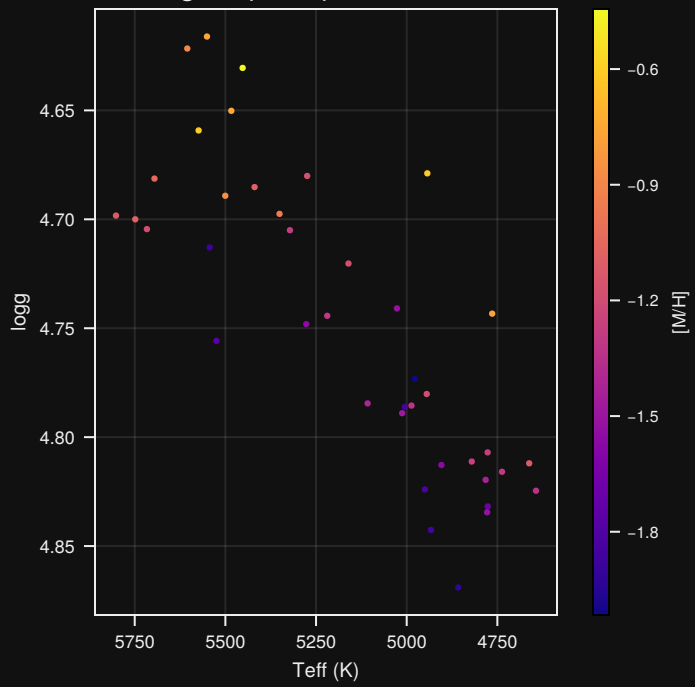
G0 & GSP-Phot coverage



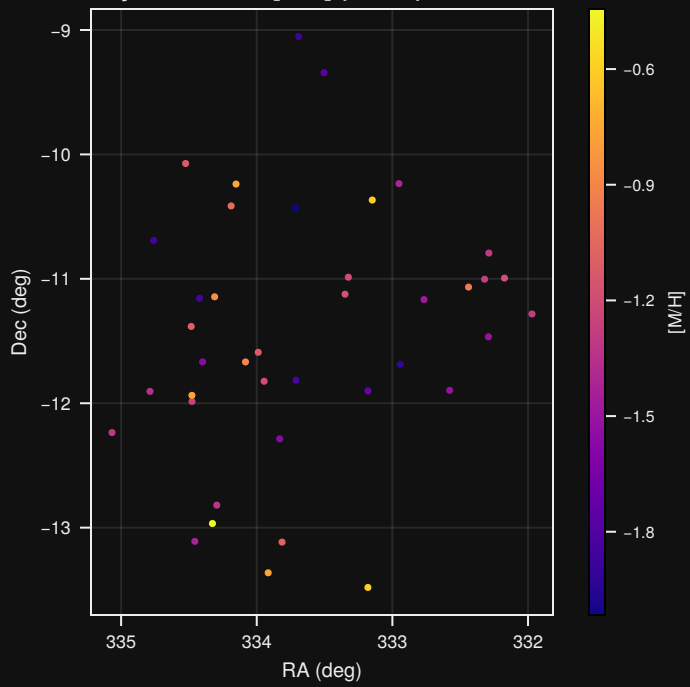
Star types from logg (n = 40)



Kiel diagram (n = 40)



Sky · GSP-Phot [M/H] (n = 40)

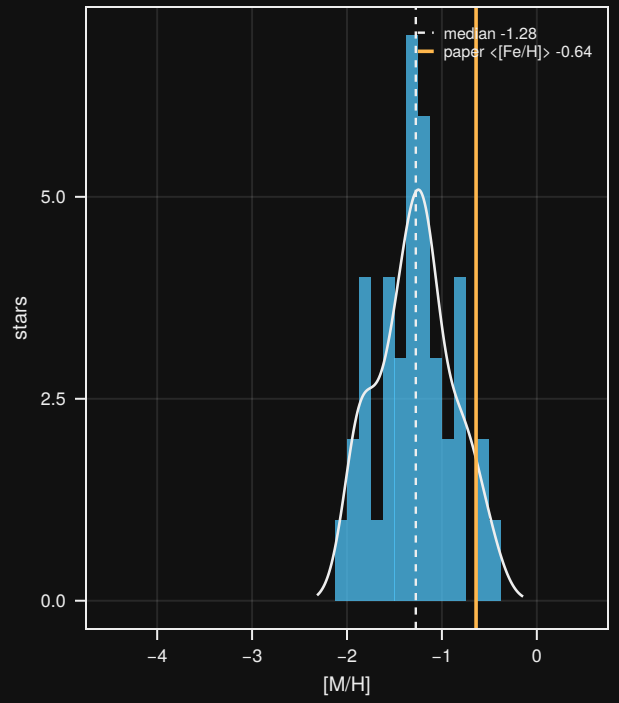


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

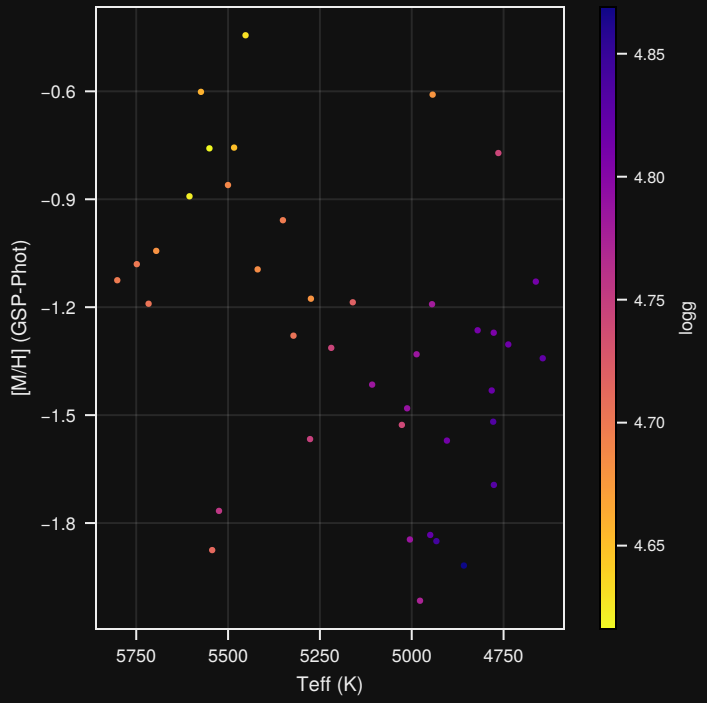
New-24

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 40)



Teff vs [M/H] (n = 40)



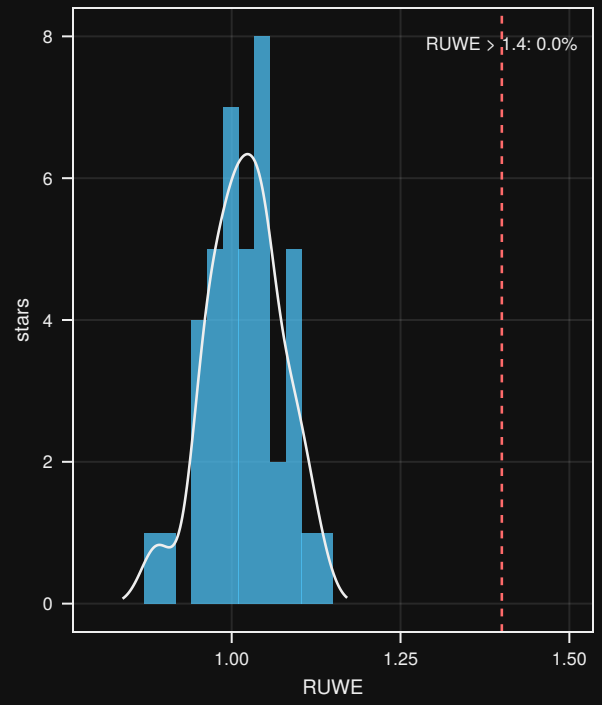
FLAME age (n = 0)

FLAME ages for only 0 star(s)
(FLAME needs good parallaxes)

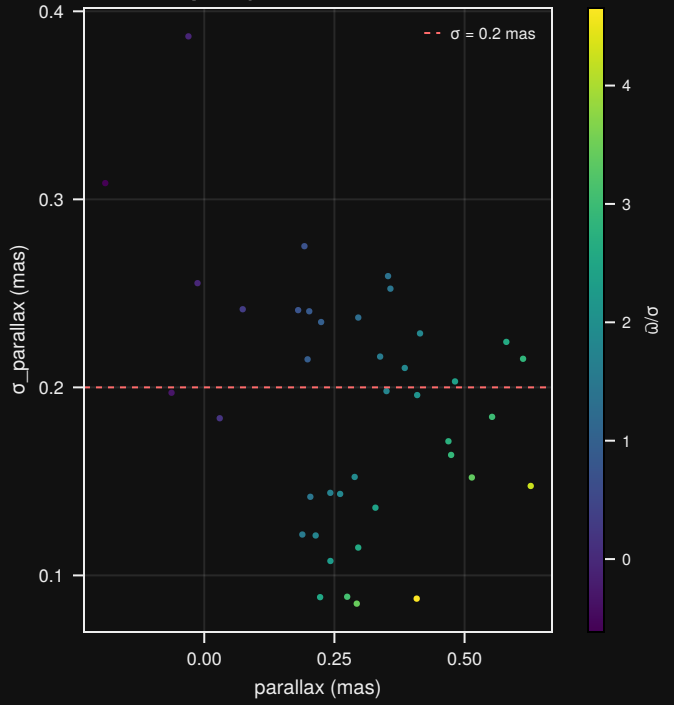
Age–metallicity (n = 0)

n < 3

RUWE (n = 40)

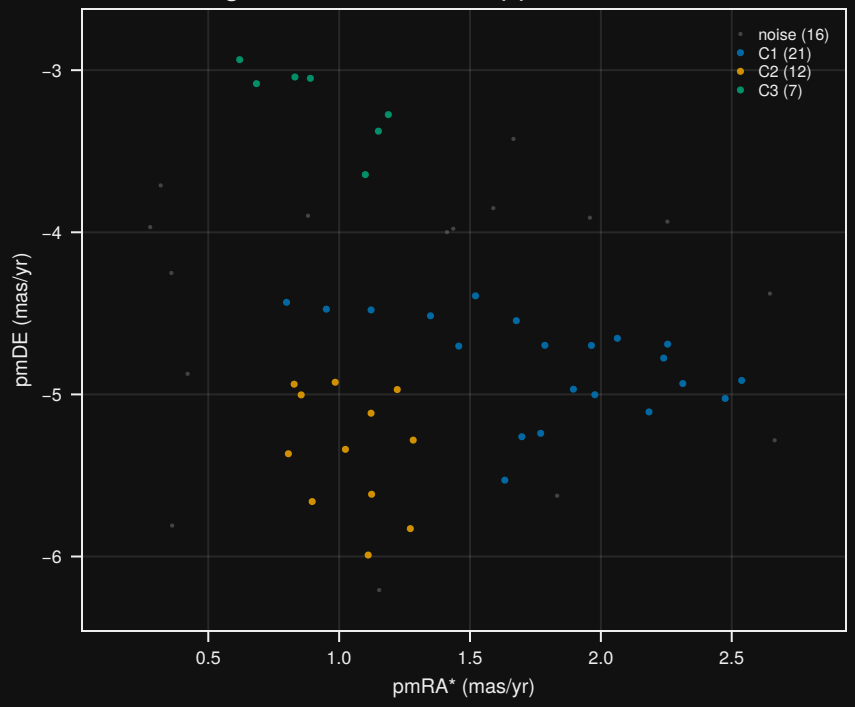


Parallax quality (n = 40)

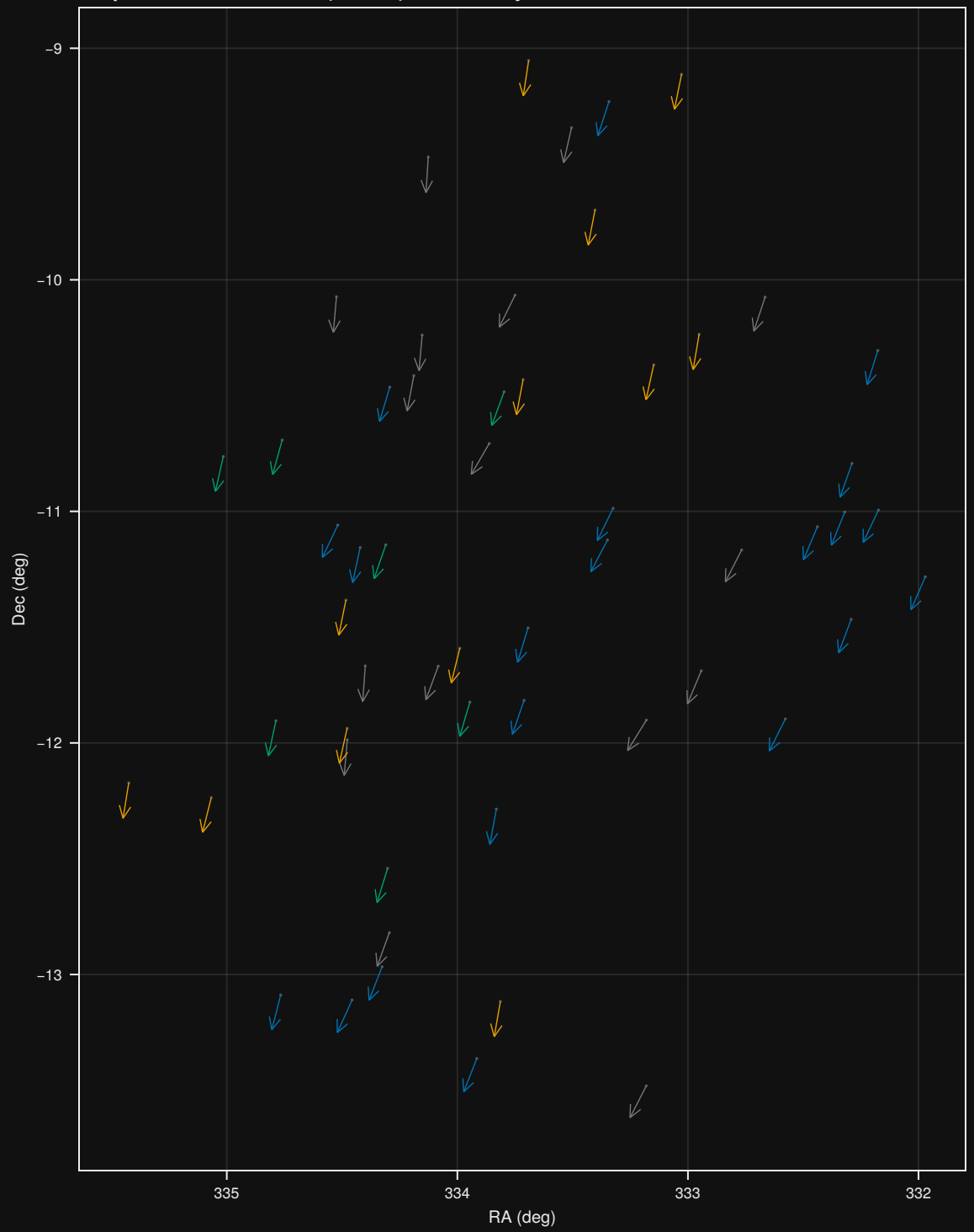


New-24

PM clustering · HDBSCAN · 3 cluster(s) · noise 29%



Proper-motion directions (arrows) coloured by cluster



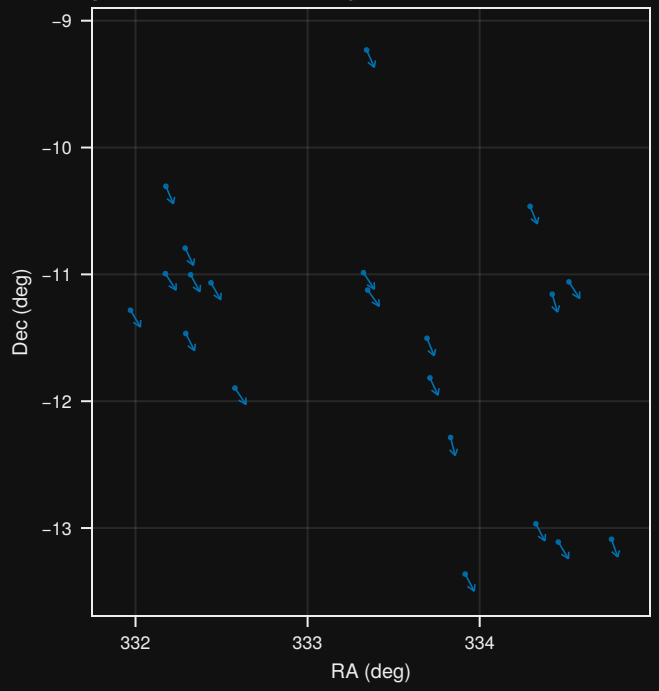
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	16	1.42	-3.99	1.02	-1.30	2
C1	21	1.79	-4.70	0.49	-1.28	2
C2	12	1.07	-5.31	0.24	-1.13	2
C3	7	0.89	-3.08	0.31	-1.27	1

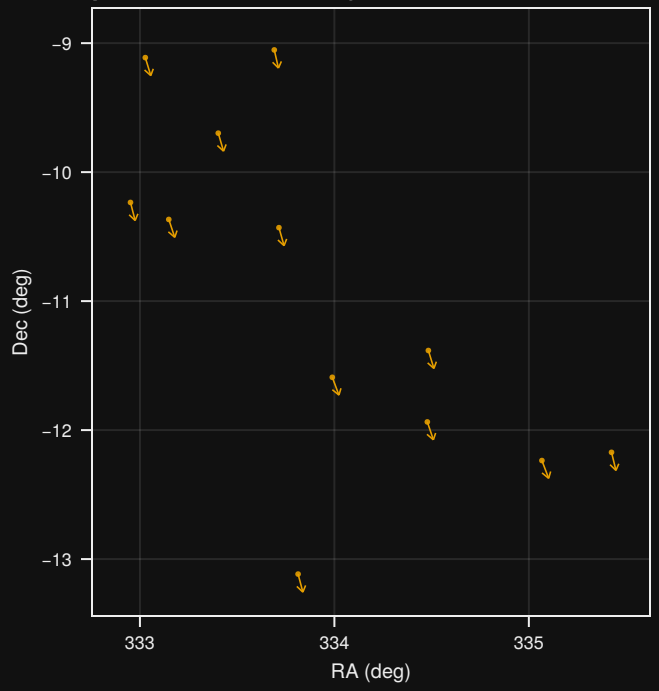
New-24

Per-cluster proper-motion vector maps

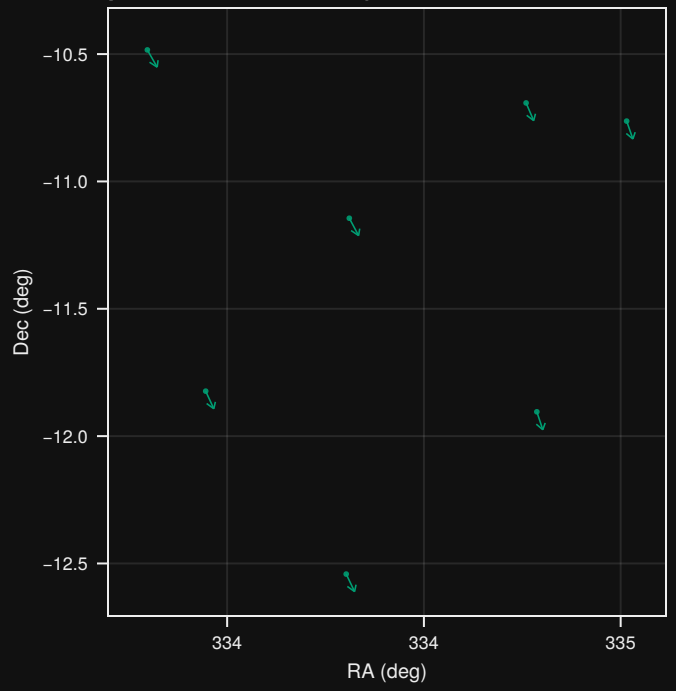
Cluster C1 · n = 21
pm = (1.79, -4.70) mas/yr



Cluster C2 · n = 12
pm = (1.07, -5.31) mas/yr



Cluster C3 · n = 7
pm = (0.89, -3.08) mas/yr



Noise · n = 16

